

EXTENSION TRAINING CENTRE (ETC) MALANGPORA PULWAMA

Sher-e-Kashmir University of Agricultural Sciences & Technology of Kashmir (SKUAST-Kashmir)

COMPLETE TECHNICAL ARCHITECTURE & DEVELOPER HANDBOOK

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1. Executive Overview & System Purpose

Background, institutional mandate, and target users

This document serves as the comprehensive engineering and development manual for the official web portal of Extension Training Centre (ETC) Malangpora, Pulwama, under SKUAST-Kashmir. The application is designed to handle student trainee lifecycle management, online ID card generation, digital admit cards, course completion certificates, academic catalogs, public announcements, and security audit logging.

The portal operates on modern web standards with an offline-first architectural model: all trainee operations function seamlessly using client-side caching while automatically synchronizing records with a centralized Supabase cloud PostgreSQL database when network connectivity is available.

2. Full Technology Stack & Library Matrix

Complete architectural composition of frontend, backend, and data tiers

Component / Tier	Technology / Tool	Version	Role & Implementation Purpose
Frontend Framework	React & TypeScript	19.x / 5.8	Component-driven UI, state management, hooks
Styling & Theme	Tailwind CSS	v4.x	Utility-first styling, official gold/green branding
Build & Dev Tool	Vite	v6.x	Hot Module Reloading (HMR), tree-shaken bundler
Backend & Server	Node.js + Express	20.x / 4.x	Custom server (server.ts) serving APIs & SPA assets
Database (Cloud)	Supabase (PostgreSQL)	v2.x SDK	Central cloud relational storage for student rosters
Database (Local)	HTML5 LocalStorage	Standard	Instant fallback offline storage and zero-latency cache
Vector Icons	Lucide React	Latest	Institutional iconography across headers & navigation
PDF Generation	jsPDF & html2canvas	2.5.2 / 1.4	Vector CR80 ID cards, admit cards & certificates
Barcode & QR	QRCode Library	1.5.4	Instant student digital verification QR codes
UI Transitions	Motion (Framer)	Latest	Accessible route fade and interactive modal animations

3. Complete Project Directory Layout

Annotated source code map for rapid navigation in code editors

The repository is modularly structured to maintain clear separation of concerns between presentation, data, external cloud services, and static public assets:

```
etc-pulwama-portal/
% % %   p u b l i c /                               #   S t a t i c   p u b l i c   f i l e s   &
%       % % %   d r _ j a v e e d _ p o r t r a i t . j p g       #   O f f i c i a l   p h o t o   o f   P r o f
%       % % %   d r _ a j a z _ p o r t r a i t . j p g         #   O f f i c i a l   p h o t o   o f   D r .
%       % % %   j a h a n g i r _ p o r t r a i t . j p g       #   O f f i c i a l   p h o t o   o f   M r .
%       % % %   b l a n k - a v a t a r . s v g                 #   N e u t r a l   s i l h o u e t t e   p l a
%       % % %   s k u a s t - k a s h m i r - l o g o . p n g     #   O f f i c i a l   h i g h - r e s o l u t i
% % %   s r c /                                             #   T y p e S c r i p t   R e a c t   A p p l i
%       % % %   c o m p o n e n t s /                         #   M o d u l a r   U I   w i d g e t s   a n d
%       %       % % %   H e a d e r . t s x                   #   S t i c k y   i n s t i t u t i o n a l   n
%       %       % % %   F o o t e r . t s x                   #   O f f i c i a l   c o n t a c t   d i r e c
%       %       % % %   S t u d e n t I d C a r d . t s x       #   I S O   C R 8 0   S t u d e n t   I D   c a
%       %       % % %   A d m i n D a s h b o a r d V i e w . t s x #   A n a l y t i c s ,   b a t c h   m e t r i
%       %       % % %   A d m i n A u d i t L o g s V i e w . t s x #   S e c u r i t y   a u d i t   t r a i l s
%       %       % % %   B u l k A n n o u n c e m e n t V i e w . t s x #   S M S / E m a i l   n o t i f i c a t i o
%       %       % % %   S u p a b a s e C o n f i g G u i d e . t s x #   I n - p o r t a l   S Q L   s c h e m a   &
%       % % %   d a t a /                                     #   S t r u c t u r e d   m o c k   r e c o r d
%       %       % % %   m o c k D a t a . t s                  #   F a c u l t y   m e m b e r s ,   c o u r s
%       % % %   l i b /                                       #   U t i l i t y   e n g i n e s ,   f o r m a
%       %       % % %   s u p a b a s e . t s                  #   S u p a b a s e   P o s t g r e s S Q L   c l
%       %       % % %   p d f G e n e r a t o r . t s          #   j s P D F   e n g i n e   f o r   e x a m i
%       %       % % %   i d C a r d P r i n t . t s            #   H i g h - r e s o l u t i o n   p r i n t
%       % % %   p a g e s /                                   #   F u l l - p a g e   r o u t e   v i e w s
%       %       % % %   H o m e P a g e . t s x               #   I n s t i t u t i o n a l   h o m e   p a g
%       %       % % %   A b o u t P a g e . t s x             #   K V K   P u l w a m a   V i s i o n ,   M i
%       %       % % %   S t u d e n t I d C a r d P a g e . t s x #   S t u d e n t   i d e n t i t y   v e r i f
%       %       % % %   D o w n l o a d s P a g e . t s x      #   O f f i c i a l   d o c u m e n t ,   s y l
%       %       % % %   C o n t a c t P a g e . t s x          #   C a m p u s   l o c a t i o n   m a p ,   c
%       %       % % %   A d m i n P a g e . t s x             #   A d m i n i s t r a t o r   c o n t r o l
%       % % %   t y p e s . t s                               #   G l o b a l   T y p e S c r i p t   d e f i
%       % % %   A p p . t s x                                 #   P r i m a r y   r o u t i n g   c o m p o n
%       % % %   m a i n . t s x                               #   R e a c t   D O M   r o o t   b o o t s t r
%       % % %   i n d e x . c s s                             #   T a i l w i n d   C S S   i m p o r t   d e
% % %   s e r v e r . t s                                     #   E x p r e s s   s e r v e r   e n t r y   p
% % %   p a c k a g e . j s o n                               #   P r o j e c t   d e p e n d e n c i e s   a
% % %   . e n v . e x a m p l e                               #   E n v i r o n m e n t   v a r i a b l e s
```

4. Local Development Setup in VS Code

Step-by-step instructions to run and modify the portal on your PC

- **Step 1: Install Node.js:** Ensure Node.js (v20.x or higher) is installed. Download from <https://nodejs.org>. Verify by running "node -v" in Command Prompt.
- **Step 2: Export Codebase:** In Google AI Studio, click Settings/Options in the top right, then select "Export to ZIP" (or push to your GitHub repo).
- **Step 3: Open in VS Code:** Extract the ZIP file to your preferred folder (e.g. C:\Projects\etc-pulwama-portal). In VS Code, click File -> Open Folder.
- **Step 4: Open Terminal:** In VS Code, press Ctrl + ` (backtick) or go to Terminal -> New Terminal.
- **Step 5: Install Dependencies:** In the terminal, execute: "npm install". This automatically downloads all libraries into the node_modules folder.

5. Environment Configuration & Secrets

Configuring cloud database credentials safely

Create a file named `.env` in the root directory (alongside `package.json`) and configure your Supabase project keys:

```
# .env configuration for local execution
PORT=3000
VITE_SUPABASE_URL=https://ssypyegksjrpjgbcoqyc.supabase.co
VITE_SUPABASE_ANON_KEY=sb_publishable_P19UWTAtI4Ujeg9HrYohqA_s6podg89
```

To start the application in development mode with live Hot Module Reloading:

```
npm run dev
```

Open your web browser and navigate to: `http://localhost:3000`. Any changes you save in VS Code will immediately update the live preview without reloading the page!

6. Supabase Cloud Database SQL Schema

Table schema and database structure for student rosters

The portal utilizes Supabase (PostgreSQL) to store student registrations. To provision the table in your own Supabase project:

- **Navigate to SQL Editor:** Log in to `https://supabase.com`, open your project, and click the "SQL Editor" tab on the left navigation.
- **Execute DDL Query:** Paste and run the following standard SQL script:

```
-- Supabase PostgreSQL Table for ETC Pulwama Trainee Roster
CREATE TABLE IF NOT EXISTS public.students (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  roll_number TEXT UNIQUE NOT NULL,
  registration_number TEXT,
  name TEXT NOT NULL,
  father_name TEXT,
  course_type TEXT,
  course_title TEXT,
  session TEXT,
  division TEXT,
  cgpa TEXT,
  email TEXT,
  phone TEXT,
  blood_group TEXT,
  dob TEXT,
  district TEXT,
  address TEXT,
  photo_url TEXT,
  status TEXT DEFAULT 'active'
);

-- Enable Row Level Security (RLS)
ALTER TABLE public.students ENABLE ROW LEVEL SECURITY;

-- Public Read & Write Policies for Portal Operations
CREATE POLICY "Allow public read access" ON public.students FOR SELECT USING (true);
CREATE POLICY "Allow public insert/update access" ON public.students FOR ALL USING (true);
```

7. Essential NPM Commands Quick-Reference

Commands for building, checking, and deploying the codebase

Command	Execution Action & Context
npm run dev	Starts Express server + Vite development server at http://localhost:3000 with HMR.
npm run lint	Runs TypeScript compiler check (tsc --noEmit) to detect syntax or type errors.
npm run build	Compiles production client bundle into dist/ and bundles server.ts via esbuild.
npm run start	Launches the compiled production server (node dist/server.cjs) for deployment.

8. Developer Customization Guide: Where to Edit What

Instant reference for common updates requested by university administration

What you want to change	Target Source File to Edit	Specific Instructions
Faculty Directory & Team	src/data/mockData.ts	Modify FACULTY_LIST array with names, designations & photo URLs.
Vision, Mission, Objectives	src/pages/AboutPage.tsx	Update KVK Pulwama Vision & Mission text blocks directly.
Official Contact & Phone	src/data/mockData.ts	Edit INSTITUTION_INFO object (phone, email: pcpulwama@gmail.com).
Courses & Syllabi	src/data/mockData.ts	Update COURSES array with duration, seats, eligibility & modules.
Flash Notices & Marquee	src/data/mockData.ts	Add or update items in the NOTICES array with priority levels.
Student ID Card Design	src/components/StudentIdCard.tsx	Tune dimensions, ISO CR80 PVC formatting, and barcode layouts.
Admit Card PDF Engine	src/lib/pdfGenerator.ts	Customize examination hall instructions, watermarks, and seals.
Admin Credentials	src/pages/AdminPage.tsx	Update admin password validation (default: admin123).

9. Frequently Asked Questions by Developers

Solutions for common development scenarios

- **Q: How do I change faculty photos?:** Place the new image inside the public/ folder (e.g. public/new_photo.jpg). Then open src/data/mockData.ts, find the faculty member in FACULTY_LIST, and set image: "/new_photo.jpg".
- **Q: How does student data persist?:** When an administrator or student saves a profile, the portal saves it instantly in localStorage (ensuring it works offline) and attempts to push it to the Supabase students table. In the Admin Panel, clicking "Sync to Supabase" synchronizes all records in bulk.
- **Q: How do I deploy to a web host?:** Run "npm run build". The output folder "dist" contains the static website files. You can deploy it to Vercel, Netlify, Cloud Run, or any standard Linux VPS running Node.js.

10. Technical Support & Institutional Verification

Official contacts for technical assistance and web portal maintenance

Extension Training Centre (ETC) Malangpora Pulwama
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- Campus Address: Koil Road, Near District Administrative Complex, Pulwama, J&K - 192301
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